Full Marks: 35

Numerical Optimization

Assignment 1

Question 1 (2 marks)

For a function $f : \mathbb{R}^n \to \mathbb{R}$, what is the difference between a local minimizer and a global minimizer?

Question 2 (4 marks)

Verify whether the following functions are convex or not

i) $f : \mathbb{R} \to \mathbb{R}$, defined by $f(x) = e^x$.

ii) $f : \mathbb{R}^2 \to \mathbb{R}$, defined by $f(x, y) = x^2 + 4xy + y^2 + x - y$.

Question 3 (4 marks)

Verify whether the following matrices are positive semidefinite or not. Justify your answer.

- $A = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 3 & 2 \\ 1 & 2 & 3 \end{pmatrix}$
- $A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 3 & 9 \end{pmatrix}$

Question 4 (2 marks)

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable convex function. If $\nabla f(x) = 0$, then prove that x is a global minimizer of f.

Question 5 (3 marks)

Suppose you want to solve the problem

$$\text{minimize } (x - 2)^2 + (x - 2y)^2.$$

You want to use the steepest decent algorithm with constant step length 0.3. Currently, you are at the point $(0, 3)$. Find the next iteration point.

Question 6 (4 marks)

Find local/global minimizers of the following functions. Justify why they are local/global minimizers. (Also mention if it doesn't have a local/global minimizer)

- $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x, y) = 8x + 12y + x^2 - 2y^2$
- $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x, y) = 100(y - x^2)^2 + (1 - x)^2$

Question 7 (8 marks)

- Suppose four data points $\{(x_i, y_i) \in \mathbb{R}^2 \mid i = 1, 2, 3, 4\}$ have been given in the plane. We want to find a quadratic line $y = ax^2 + bx + c$ such that $\sum_{i=1}^{4}(y_i - ax_i^2 - bx_i - c)^2$ is minimized. Write it as a least square problem (that is, express it as a quadratic problem by calculating the quadratic defining matrix, etc).
- For the particular choice of the following points, find the best-fitting quadratic curve in the least square sense using Python code:

  i) $\{(1, 10),\ (-1, 4),\ (3, 32),\ (-2, 7)\}$

  ii) $\{(1, 10),\ (-1, 5),\ (2, 10),\ (3, 15)\}$

Question 8 (2 marks)

Will pure Newton's method be successful if we want to minimize $f(x_1, x_2) = x_1^3 + x_1x_2 - x_1^2x_2^2$ starting from the point $(1, 1)$? Justify your answer.

Question 9 (2 marks)

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x_1, x_2) = 2x_1^2 - 3x_2^2$. Is it a function with a Lipschitz gradient?

Question 10 (4 marks)

Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) = x_1^2 + 3x_1x_2 + x_2^2$. Find two distinct (linearly independent) decent directions of this function at the point $(1, 1)$. Also, along any one of these directions, find a step length that satisfies Armijo's condition with $c_1 = 0.75$.